



PERSONNEL DEPARTMENT

Official T-Shirt Production Specification

Crew-neck tee · Military green · DTF heat transfer · Sizes S–3XL

MANDATORY PRODUCTION STANDARD
Binding on every production run once signed in Section 14

Document number	_____
Revision / date	Rev 0 · 6 October 2026
Insignia-use approval reference	_____
Approving authority	_____
Order / purchase reference	_____
Authorized vendor	_____
Authorized quantity (all sizes)	_____

1 Purpose, scope and authority

This document sets out how the official Personnel Department crew-neck T-shirt must be produced: garment, artwork, crest placement, dimensions, print process, quality control, insignia etiquette and disposal of rejects. It applies to every vendor, printer and person who handles the artwork or produces the garment.

- **Mandatory.** Every requirement written as "must" or "never" is compulsory. Any deviation needs prior written approval from the approving authority named on the cover.
- **Order of precedence.** If documents conflict, the order is: (1) the signed approved pre-production sample, (2) this specification, (3) the master artwork files, (4) the purchase order.
- **Authorized use only.** The crest is an official insignia. It is authorized for this order and this quantity only, under the approval reference on the cover.
- Values marked **†** are recommended defaults set for this specification. The approving authority may change them before signing.

2 Definitions and measuring method

Term	Meaning
HPS (high point shoulder)	The point where the shoulder seam meets the collar rib, on the wearer's left side.
CF (centre front)	The vertical centre line of the garment front, found by folding the garment in half lengthwise, sleeve to sleeve.
Front collar seam	The stitch line where the collar rib joins the front body at CF.
Wearer's left	The left side of the person wearing the shirt. On a garment laid face-up, this is the right side as you look at it.
Print width	The horizontal distance between the outermost printed points of the artwork (the ends of the Khmer text line).
PPS	Pre-production sample: a finished garment made with the bulk materials and process, submitted for written approval before bulk production.

How to measure: lay the garment flat, face-up, smooth and unstretched, on a hard surface. Measure with a steel rule or a calibrated tape, in centimetres. Find CF by folding, never by eye.

3 Garment specification

Item	Requirement
Style	Short-sleeve crew-neck T-shirt, regular fit, rib collar, side-seamed or tubular body
Fabric	100% cotton single jersey, combed or ring-spun, pre-shrunk; 180–200 g/m ² †
Colour	Military green, matched to the approved physical colour standard. Digital reference ≈  #63684B (sampled from the reference image; for screens only). Textile code (e.g. Pantone TCX): _____
Size range	S, M, L, XL, 2XL, 3XL. Reference size M: chest width 51 cm, measured flat 2.5 cm below the armhole
Shade consistency	All garments in one order must come from the same dye lot wherever possible. Mixed lots must be packed separately and declared.
Labels	Size and care label inside the neck or side seam only. No vendor or commercial branding on the outside of the garment.

4 Approved artwork and master files

Production must use **only** the master files below, exactly as supplied. They are transparent PNGs at 300 dpi, sRGB, with hard edges: every pixel is fully opaque or fully transparent, with no soft shadows, so the white underbase prints clean.

File	Use	Pixels (W × H)	Print size (W × H)
front_S_25cm.png	Full front, size S	2953 × 2888	25.0 × 24.5 cm
front_M-L_28cm.png	Full front, sizes M and L	3307 × 3234	28.0 × 27.4 cm
front_XL-2XL_30cm.png	Full front, sizes XL and 2XL	3543 × 3464	30.0 × 29.3 cm
front_3XL_32cm.png	Full front, size 3XL	3780 × 3696	32.0 × 31.3 cm
leftchest_10cm.png	Left chest, all sizes	1181 × 1155	10.0 × 9.8 cm

Never redraw, trace, re-type, re-colour, or regenerate the crest or its Khmer text, including with AI image tools. Never take the crest from the internet, social media or a photograph. If a file seems damaged or the wrong size, stop and ask for a replacement master file.

Approved wording. The Khmer text must match the master file exactly: banner អង្គបញ្ជាការ; title line ទីបាត់ការបុគ្គលិក.

5 Colour reference

These values were sampled from the master artwork. Use them to check colour on screen and in the RIP. The **approved PPS under daylight (D65)** is the final colour standard.

Element	Reference (sRGB)	Notes
Gold, highlight	 #FBE492	Metallic gold is printed as a CMYK gradient, not foil. Keep the gradient; do not flatten it to one gold.
Gold, mid-tone	 #EFCB61 / #D5AC5E	
Gold, shade	 #A46C2F	
Royal navy (shield, banner, text outline)	 #131A63	Must stay distinctly blue, not black
Red (star)	 #A31917	Must not shift towards orange or brown
Khmer title fill	 #FDDB6A	Bright yellow-gold. Must be clearly legible on military green.

6 Logo placement



Figure 1. Placement template, reference size M (scale set by chest width 51 cm).

Position	Artwork	Vertical reference	Horizontal reference
A · Full front	front_[size].png	Top edge of the artwork 7.5 cm below the front collar seam	Centred on CF
B · Left chest (wearer's left)	leftchest_10cm.png	Centre of the artwork 18 cm below HPS	Centre of the artwork 10 cm from CF, towards the wearer's left
C · Back yoke	Reserved. Width 7–8 cm, top edge 2.5–3.5 cm below the back collar seam. Not authorized until separate artwork is approved and issued.		

A and B are alternatives: one garment carries either A or B, never both, unless the purchase order states otherwise. Purchase order position(s): _____

Placement tolerances [†]

Check	Tolerance
Vertical position (A: top edge; B: centre)	± 0.5 cm
Horizontal position (A: centring on CF; B: 10 cm offset)	± 0.3 cm
Tilt: the Khmer text baseline must be parallel to the hem	≤ 1°
Print width compared with the specified width	± 2 %



Figure 2. Position A, full front (size M, 28 cm).



Figure 3. Position B, left chest (10 cm).

7 Size and dimension chart

Garment size	Position A file	A print size (W × H)	A top edge below collar seam	B print size	B centre (from CF / below HPS)
S	front_S_25cm	25.0 × 24.5 cm	7.5 cm	10.0 × 9.8 cm	10 / 18 cm
M	front_M-L_28cm	28.0 × 27.4 cm	7.5 cm	10.0 × 9.8 cm	10 / 18 cm
L	front_M-L_28cm	28.0 × 27.4 cm	7.5 cm	10.0 × 9.8 cm	10 / 18 cm
XL	front_XL-2XL_30cm	30.0 × 29.3 cm	7.5 cm	10.0 × 9.8 cm	10 / 18 cm
2XL	front_XL-2XL_30cm	30.0 × 29.3 cm	7.5 cm	10.0 × 9.8 cm	10 / 18 cm
3XL	front_3XL_32cm	32.0 × 31.3 cm	7.5 cm	10.0 × 9.8 cm	10 / 18 cm

- Maximum artwork height at position A is **33 cm**. Every file above is within this limit.
- Scale the artwork **proportionally only**. Width and height must always stay locked together.
- The size-M placement figures apply to every size. Check placement on the S and 3XL samples during PPS approval (Section 11). Any change for a particular size must be agreed in writing and added to this table.
- Minimum detail: printed lines must be at least 0.5 mm wide [†]. At 10 cm (position B), the banner text អង្គបញ្ជាការ is about 3 mm tall. Its legibility must be confirmed on the PPS before bulk production.

8 Insignia and design etiquette

The crest represents the institution and must be treated with respect at every stage, from file handling to finished garment. Everyone involved in production must follow these rules.

1. **Reproduce exactly.** Use the master files only, at 100% of their supplied proportions.
2. **Never alter the crest.** Do not re-colour, stretch, squash, rotate, mirror, crop or partially hide it. Do not add or remove elements, and do not add effects such as shadow, glow, bevel or distressing.
3. **Upright, always.** The crown and radiant star must point straight up towards the collar.
4. **Position of honour.** The crest may only be printed on the upper torso, at the approved positions: full front centred, or left chest over the heart. Never below the waist, on sleeves, at the hem, or anywhere it could be sat on or folded under.
5. **Clear space.** Keep a clear zone of at least 10% of the print width on every side (2.8 cm at 28 cm, 1.0 cm at 10 cm). No other graphics, text, seams or pockets may enter it.
6. **Precedence.** No other logo, slogan or graphic may be printed larger than the crest, above it, or touching it. No commercial, vendor or sponsor marks on the outside of the garment.
7. **Approved background only.** Print only on the approved military green. Any other garment colour needs new written approval and a new PPS.
8. **Exact wording.** Never re-type or re-set the Khmer text in another font. It is part of the approved artwork.
9. **Respectful handling.** Printed garments and transfer films must not be placed on the floor, stepped on or used as wipes. Store them flat and covered.

Clear space



The dashed line marks the outer edge of the artwork. The clear zone around it is at least 10% of the print width on every side. Nothing may be placed inside it.

Correct and incorrect use

✓ Correct Master file, upright, approved green	✗ Stretched or squashed Proportions must stay locked	✗ Rotated or tilted Always upright	✗ Re-coloured No colour changes
✗ Cropped Show the complete crest and text	✗ Effects added No shadow, glow or bevel	✗ Unapproved garment colour Military green only	✗ Inverted or mirrored Critical defect

9 File preparation by software

Use any of the applications below to check the files, lay them out or hand them to the RIP. In every application the output must be a **transparent PNG** at the pixel size in Section 4, sRGB, with nothing added.

Application	Canvas / document setup	Placing and exporting
Adobe Photoshop	File > New: width and height in cm from Section 4; Resolution 300 px/inch; RGB, 8-bit; Background Contents: Transparent	File > Place Embedded, then confirm 100% scale (no resampling). File > Export > Export As > PNG: Transparency on, "Smaller file (8-bit)" off, scale 100%. Check Image > Image Size shows the Section 4 pixel size.
Adobe Illustrator	Artboard in cm at the Section 4 print size; RGB document	File > Place (linked). In the Links panel, effective PPI must read 300. File > Export > Export As > PNG: Resolution 300 ppi, Background Transparent, Anti-aliasing None (keeps the hard edges).
Adobe Express	Custom size in pixels (e.g. 3307 × 3234 px for M/L)	Upload the PNG and size it to fill the canvas exactly. Download > PNG with Transparent background. Check the downloaded file's pixel size.
Figma	Frame at the Section 4 pixel size; remove the frame fill so it is transparent	Place the image at W/H exactly equal to the frame, with no crop. Export > PNG > 1x. Figma reduces images over 4096 px, but all master files are under that. Figma does not save DPI, so the RIP operator must set the print width from Section 7.
Canva	Create design > Custom size in px (not cm), Section 4 pixel size	Upload, then set the image to fill the page with no crop. Share > Download > PNG, size ×1, tick Transparent background. Do not apply Canva filters or "Enhance". Check the downloaded file's pixel size.
Sketch	Artboard at the Section 4 pixel size; untick Background Color	Insert the image at exact W/H. Export > PNG > 1x. Like Figma, the RIP operator sets the print width.

In every application, never: export as JPG; flatten onto a white or green background; apply automatic enhancement, sharpening, compression or "remove background"; upscale beyond the master files; or embed the artwork in a mockup and print the mockup.

10 DTF print and press process

10.1 Printing (RIP)

- Import the master PNG and set the print width from Section 7. Lock the aspect ratio, and do not resample beyond the master file.
- Colour: sRGB input, perceptual or relative colorimetric rendering. Match the approved PPS, not the screen.
- White underbase: 100% under all printed areas, choked by 1–2 px (0.1–0.2 mm) [†], with no white outside the artwork.
- Print a 10 cm test strip at the start of every session and compare it with the approved PPS.

10.2 Powder and curing

- Apply the hot-melt powder evenly, remove all excess, and cure at the film and powder supplier's specified temperature and time. Record the supplier, product and batch in the production log.

10.3 Heat press

Step	Temperature	Time	Pressure / notes
Pre-press garment (removes moisture and wrinkles)	160 °C	3–5 s [†]	Light
Transfer press (cotton)	160 °C	15 s	Medium-firm; peel hot or cold according to the film supplier's instructions
Finishing press	160 °C	10 s	Medium-firm, with a PTFE (Teflon) sheet or parchment over the print

Align with a T-square or laser guide using the measurements in Sections 6 and 7. Never align by eye.

10.4 Care instructions (on the care label)

Wash inside out at a maximum of 30 °C · No bleach · Do not tumble dry on high heat · Do not iron directly on the print · Do not dry-clean.

11 Quality control and acceptance

11.1 Pre-production sample (PPS)

- The vendor must submit **one PPS per position for sizes S, M and 3XL** [†], made with the bulk garment, film, ink and press settings.
- Bulk production may only start after the PPS is approved in writing. The approved PPS is then sealed, signed and kept as the reference standard.

11.2 Tests on the PPS and first bulk pieces

Test	Method	Pass criterion
Placement	Measure as in Section 2	Within the Section 6 tolerances
Colour	Side by side with the approved PPS under daylight (D65)	No visible difference in the gold, navy or red
Edge quality	Inspect with a 10× loupe	No white halo, no fuzzy edges, all fine lines intact
Stretch	Stretch the print area by hand to about 1.5× its width, hold, release	No cracking, no lifting
Wash durability	5 cycles at 30 °C, inside out, line dry	No peeling, cracking or noticeable fading
Adhesion	Try to lift the print edge with a fingernail	Edges stay fully bonded

11.3 Bulk inspection

Inspect each lot to ISO 2859-1, general inspection level II, using these acceptance quality limits [†]: **Critical 0** · **Major 2.5** · **Minor 4.0**.

Class	Defects
CRITICAL	Altered, distorted or incomplete crest; misspelled or re-typed Khmer text; crest inverted, mirrored or tilted; crest printed at an unapproved position; wrong garment colour; any foreign mark on the garment. Zero accepted. Garments with a critical defect must be destroyed (Section 13).
MAJOR	Placement outside tolerance; visible colour shift from the PPS; white halo or underbase showing; peeling, cracking or bubbles; ghosting or double image; smudges or stains in the print area; garment hole or seam defect.
MINOR	Slight gloss variation; loose threads away from the print; small packing or labelling faults.

12 Packing, labelling and delivery

- Fold each garment with the print facing out and centred. Put tissue or a PTFE sheet over the print if it is folded before it has fully cooled.
- Individual poly bag with a size sticker. Pack cartons by size, with the size, quantity and order reference on two sides.
- Delivery note to include: quantity per size and position, number of rejects destroyed, and transfer films printed and used (Section 13 reconciliation).

13 Security, confidentiality and disposal

- Master files are confidential. Store them only on the production workstation and never share them with third parties or post them online.
- Produce only the authorized quantity. Overruns, spare transfers and spare films must be handed back to the department or destroyed under supervision.
- Rejects and misprints** must be destroyed by cutting through the crest before disposal. They must never be sold, donated or reused intact.
- After delivery, delete the master files from all vendor systems and confirm in writing.
- Reconciliation** (recorded on the delivery note): transfers printed = garments delivered + rejects destroyed + spare transfers returned or destroyed.

14 Approval and sign-off

This specification becomes mandatory for production once it is signed below.

Role	Name and rank	Signature	Date
Prepared by			
Checked by			
Approved by			

Vendor acknowledgement

We have received and read this specification and the master artwork files. We agree to produce, inspect, pack, secure and dispose of all materials exactly as specified, and to request written approval before any deviation.

Company	Authorized representative	Signature and stamp	Date

Revision history

Rev	Date	Change	Approved by
0	6 Oct 2026	First issue for approval	